網路架構
ONTAP 9

NetApp
May 08, 2022

This PDF was generated from https://docs.netapp.com/zh-tw/ontap/concepts/network-connectivity-concept.html on May 08, 2022. Always check docs.netapp.com for the latest.

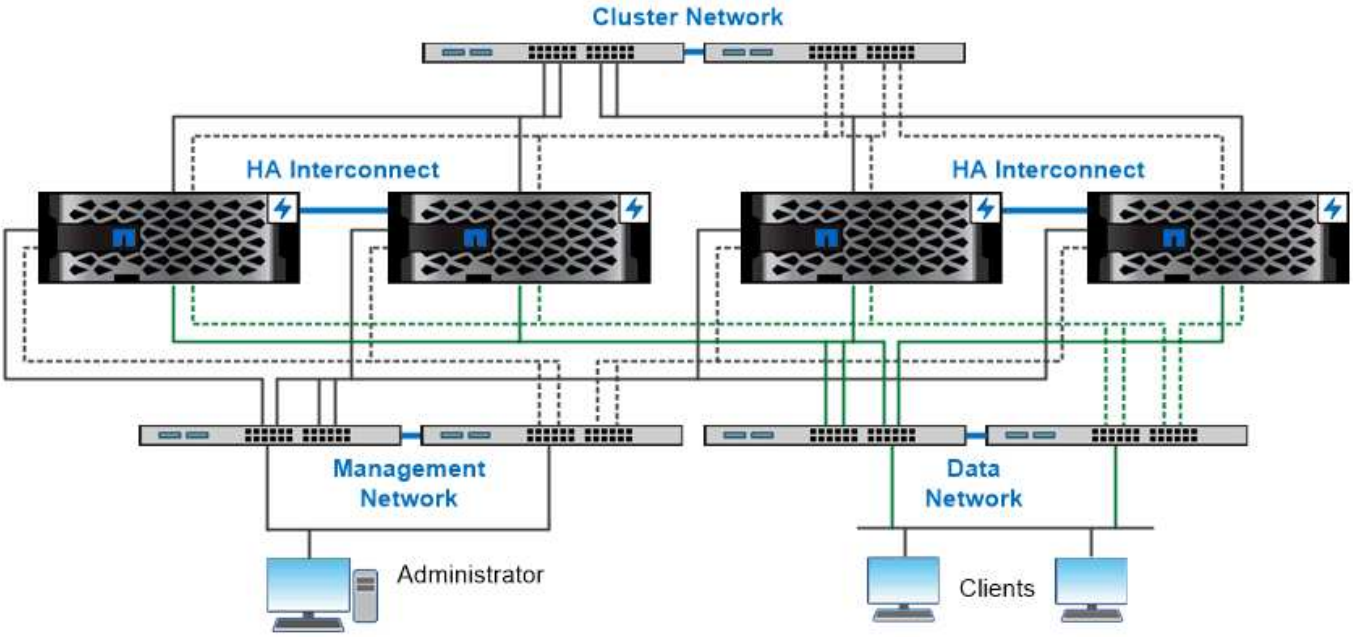
目錄

網路架構 1
網路架構總覽 1
邏輯連接埠 1
支援業界標準網路技術 2
RDMA總覽 3

網路架構

網路架構總覽

實作資料中心的網路架構ONTAP 通常包含叢集互連、叢集管理的管理網路、以及資料網路。NIC（網路介面卡）提供實體連接埠以供乙太網路連線使用。HBA（主機匯流排介面卡）提供實體連接埠以供FC連線使用。



The network architecture for an ONTAP datacenter implementation typically consists of a cluster interconnect, a management network for cluster administration, and a data network.

邏輯連接埠

除了每個節點上提供的實體連接埠之外、您也可以使用_logical連接埠_來管理網路流量。邏輯連接埠是介面群組或VLAN。

介面群組

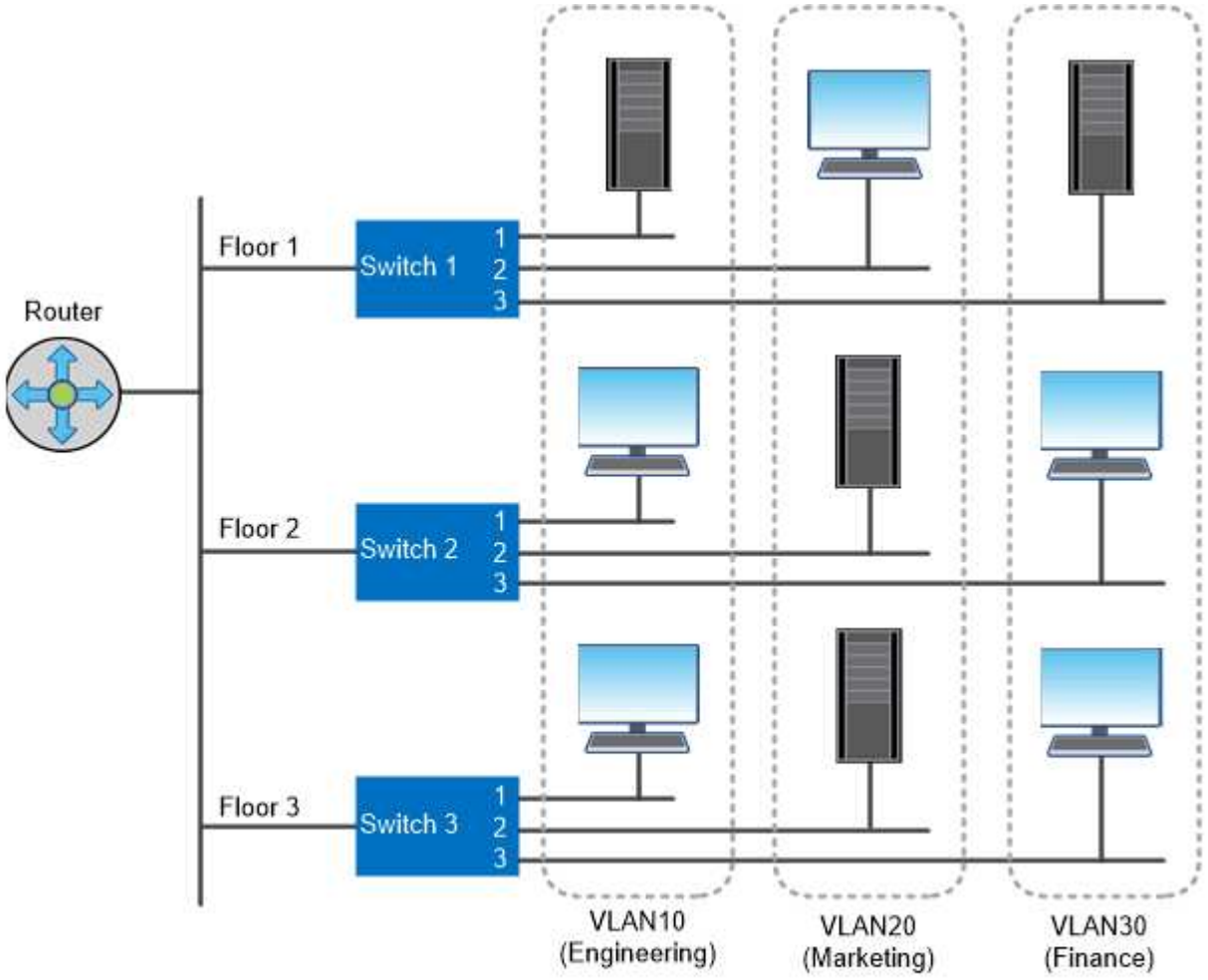
_Interface Groups_將多個實體連接埠合併成單一邏輯「'trunk port.'」 您可能想要在不同的PCI插槽中建立一個介面群組、由NIC的連接埠組成、以確保不會發生插槽故障而導致業務關鍵流量下降。

介面群組可以是單一模式、多重模式或動態多重模式。每種模式都提供不同的容錯層級。您可以使用任一類型的多重模式介面群組來平衡網路流量負載。

VLAN

_vlans_可將流量從網路連接埠（可能是介面群組）分隔成以交換器連接埠為基礎而非實體邊界所定義的邏輯區段。屬於VLAN的_end-stations是根據功能或應用程式而定。

您可以依部門（例如工程與行銷部門）或專案（例如releasease1和releasease2）來分組終端站台。由於終端站台的實體鄰近與VLAN無關、因此終端站台可在地理上處於遠端位置。



You can use VLANs to segregate traffic by department.

支援業界標準網路技術

支援所有主要的產業標準網路技術。ONTAP關鍵技術包括IPspaces、DNS負載平衡及SNMP設陷。

廣播網域、容錯移轉群組及子網路的說明請參閱 NAS路徑容錯移轉。

IPspaces

您可以使用_IPspace_為叢集中的每個虛擬資料伺服器建立不同的IP位址空間。這樣做可讓管理性分隔網路網域中的用戶端存取叢集資料、同時使用相同IP位址子網路範圍中重疊的IP位址。

例如、服務供應商可以為使用相同IP位址的租戶設定不同的IPspace、以存取叢集。

DNS負載平衡

您可以使用_DNS負載平衡_、在可用的連接埠之間分散使用者網路流量。DNS伺服器會根據介面上掛載的用戶端數目、動態選取流量的網路介面。

SNMP設陷

您可以使用_SNMP陷阱_定期檢查作業臨界值或故障。SNMP設陷會擷取從SNMP代理程式非同步傳送至SNMP管理程式的系統監控資訊。

- FIPS相容性*

所有SSL連線均符合聯邦資訊處理標準（FIPS）140-2。ONTAP您可以開啟和關閉SSL FIPS模式、全域設定SSL傳輸協定、以及關閉任何弱式加密器、例如RC4。

RDMA總覽

如果您有對延遲敏感或高頻寬的工作負載、您可能會想要善用ONTAP的讀取直接記憶體存取（RDMA）產品。RDMA可讓資料直接複製到儲存系統記憶體和主機系統記憶體之間、避免CPU中斷和負荷。

從功能組態9.10.1開始ONTAP 、您就可以進行設定 "NFS over RDMA" 在支援NVIDIA GPU的主機上、啟用NVIDIA GPUDirect Storage以GPU加速工作負載。

此外、針對採用Pensando叢集NIC的A400或A400儲存系統的非功能性使用者、推出RDMA叢集互連技術。ONTAP ONTAPRDMA叢集互連可減少延遲、縮短容錯移轉時間、並加速叢集中節點之間的通訊。如果已設定適當的儲存系統、則不需要額外的組態。

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system- without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.